5-MINUTE REVISION

# Design of Experiments

Treatments, units and factors, experimental error, Fisher's three principles and ways to improve precision.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

## Randomise, replicate and control local variation.

**REMEMBER**

Replication estimates experimental error; randomisation protects against systematic allocation bias; blocking reduces nuisance variation.

**APPLY IT**

Ten leaves measured on one treated plant are subsamples, not ten independently treated experimental units.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

## 1 Why design experiments?

**Design of Experiments (DoE)**: planning experiments so treatment effects are estimated **accurately, efficiently and without bias**.

**Objectives:** validity · reliability · replicability · optimality.

**A good design** gives unbiased estimates with their uncertainty, detects important differences, includes the analysis plan, is easy to interpret, has wide validity, uses minimal resources and is as simple as possible.

## 2 Key terms

| | |
|---|---|
| **Treatment** | Any object of comparison |
| **Control** | Standard treatment for comparison |
| **Experimental unit** | Smallest unit to which a treatment is **independently** applied |
| **Observational unit** | Unit on which the response is measured |
| **Response** | Measured outcome |
| **Factor / levels** | Variable studied / its settings |

## 3 Experimental error

Variation among experimental units **receiving the same treatment**, caused by **extraneous factors**.

Every design must provide a way to **estimate** it.

## 4 Fisher's three basic principles

| Principle | What it means | Why |
|---|---|---|
| **Randomization** | Treatments assigned to units purely by chance | Protects against allocation bias; balances unknown sources of variation on average |
| **Replication** | Same treatment on more than one unit | **Estimates experimental error**; improves precision and reliability |
| **Local control** (blocking) | Group similar units into blocks before randomising | **Reduces** experimental error from known variation |

In a **Randomized Block Design**, all treatments are randomly assigned within each block of similar units.

## 5 Improving precision

- ✓ **Border effect:** neighbouring plots affect border plants, so record only the inner **net plot**.
- ✓ **Proper plot technique:** keep everything except the treatment uniform.
- ✓ **Appropriate analysis:** e.g. ANCOVA adjusts for covariates.
- ✓ **Planning and recording:** block for time or batches; keep original records.
- ✓ **Matching the design** to the experimental units.

## 6 History

- **R.A. Fisher** founded the statistical principles of DoE at **Rothamsted Experimental Station** in the 1920s–30s.
- He introduced **ANOVA** to partition variation into factors and chance.
- With **Frank Yates**, he built methods still central to agricultural research.
- Fisher's **tea-tasting** experiment shows why randomization matters.

## 7 Example: pot experiment with manures

| | |
|---|---|
| **Treatments** | Poultry manure, cow dung, coir-pith compost |
| **Experimental unit** | One pot receiving one manure |
| **Replication** | Several pots per manure |
| **Response** | Yield per plant |
| **Error** | Yield differences among pots given the same manure |

## ! Common mistakes

- ✗ **Pseudo-replication:** counting several plants in one pot as replicates.
- ✗ Randomizing the field but always applying treatments in the same order.
- ✗ Recording border plants.

## ? Fill in the blanks

1. Applying each fertiliser dose to four plots uses the principle of ____. [ replication ]
2. Assigning treatments to plots by drawing lots is ____. [ randomization ]
3. Grouping plots of similar fertility into blocks applies ____. [ local control ]
4. Yield differences between plots that received the same treatment measure experimental ____. [ error ]
5. Counting three plants in one pot as three replicates is the mistake called ____. [ pseudo-replication ]
6. Border rows are left out and yield is recorded from the ____ plot. [ net ]